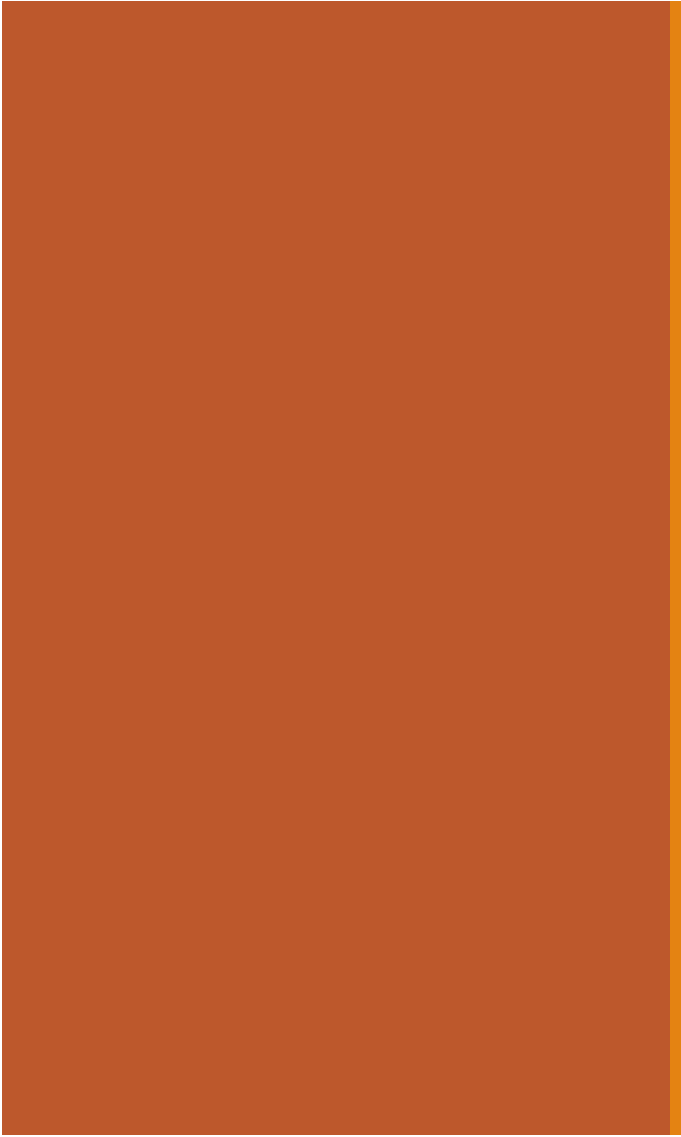


21: Bayesian Statistics and Beta

Jerry Cain
May 17, 2024

[Lecture Discussion on Ed](#)



MLE: Multinomial

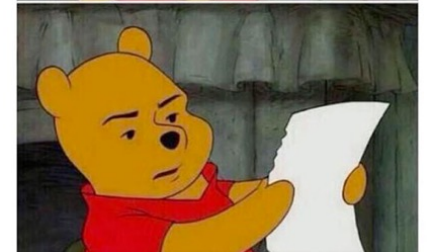
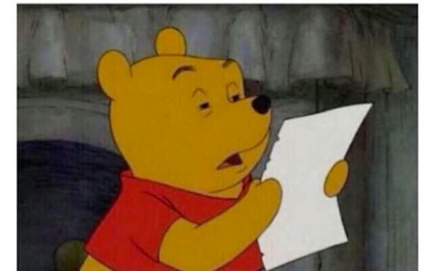
Okay, just one more MLE with the Multinomial

Consider a sample of n iid random variables where:

- Each element is drawn from one of m outcomes.
 $P(\text{outcome } i) = p_i$, where $\sum_{i=1}^m p_i = 1$
- $X_i = \#$ of trials with outcome i , where $\sum_{i=1}^m X_i = n$

*this is the classic
description of
the multinomial
distribution.*

Staring at my math homework like



Let's give an
example!

Okay, just one more MLE with the Multinomial

Consider a sample of n iid random variables where:

- Each element is drawn from one of m outcomes.
 $P(\text{outcome } i) = p_i$, where $\sum_{i=1}^m p_i = 1$
- $X_i = \#$ of trials with outcome i , where $\sum_{i=1}^m X_i = n$

m is the number of possible outcomes

p_i is the probability seeing i th of m outcomes.

Example: Suppose each RV is outcome of 6-sided die.

$$m = 6, \sum_{i=1}^6 p_i = 1$$

no assumption about fairness — we don't know $p_1, p_2, \dots$

- Roll the dice $n = 12$ times.
- Observe data: 3 ones, 2 twos, 0 threes, 3 fours, 1 fives, 3 sixes

$$\begin{aligned} X_1 &= 3, X_2 = 2, X_3 = 0, \\ X_4 &= 3, X_5 = 1, X_6 = 3 \end{aligned}$$

$$\text{Check: } \underbrace{X_1}_3 + \underbrace{X_2}_2 + \dots + \underbrace{X_6}_3 = 12$$

Okay, just one more MLE with the Multinomial

Consider a sample of n iid random variables where:

- Each element is drawn from one of m outcomes.
 $P(\text{outcome } i) = p_i$, where $\sum_{i=1}^m p_i = 1$
- $X_i = \#$ of trials with outcome i , where $\sum_{i=1}^m X_i = n$

1. What is the likelihood of observing the sample $(X_1, X_2, \dots, X_m)$, given the probabilities $p_1, p_2, \dots, p_m$?

A.
$$\frac{n!}{X_1! X_2! \dots X_m!} p_1^{X_1} p_2^{X_2} \dots p_m^{X_m}$$

B.
$$p_1^{X_1} p_2^{X_2} \dots p_m^{X_m}$$

C.
$$\frac{n!}{X_1! X_2! \dots X_m!} X_1^{p_1} X_2^{p_2} \dots X_m^{p_m}$$

$$\binom{n}{X_1, X_2, \dots, X_m} p_1^{X_1} p_2^{X_2} \dots p_m^{X_m}$$

if p_1, p_2, p_3 , etc. are unknown, then they are the unknown parameters that can, in principle, take on a range of values. We want to choose values that maximize likelihood.



Okay, just one more MLE with the Multinomial

Consider a sample of n iid random variables where:

- Each element is drawn from one of m outcomes.

$P(\text{outcome } i) = p_i$, where $\sum_{i=1}^m p_i = 1$

- $X_i = \#$ of trials with outcome i , where $\sum_{i=1}^m X_i = n$

here $\theta = (p_1, p_2, p_3, p_4, \dots, p_m)$

- What is the likelihood of observing the sample $(X_1, X_2, \dots, X_m)$, given the probabilities $p_1, p_2, \dots, p_m$?

$$L(\theta) = \frac{n!}{X_1! X_2! \dots X_m!} p_1^{X_1} p_2^{X_2} \dots p_m^{X_m}$$

- What is θ_{MLE} ?

$$LL(\theta) = \log(n!) - \sum_{i=1}^m \log(X_i!) + \sum_{i=1}^m X_i \log(p_i), \text{ such that } \sum_{i=1}^m p_i = 1$$

this constraint complicates the pure calculus approach.

Optimize with
Lagrange multipliers in
extra slides

$$\theta_{MLE}: p_i = \frac{X_i}{n}$$

Intuitively, probability
 $p_i = \text{proportion of outcomes}$

When MLEs attack!

$$\text{MLE for Multinomial: } p_i = \frac{X_i}{n}$$

Consider a 6-sided die.

- Roll the dice $n = 12$ times.
- Observe: 3 ones, 2 twos, 0 threes, 3 fours, 1 fives, 3 sixes

What is θ_{MLE} ?



When MLEs attack!

$$\text{MLE for Multinomial: } p_i = \frac{X_i}{n}$$

Consider a 6-sided die.

- Roll the dice $n = 12$ times.
- Observe: 3 ones, 2 twos, 0 threes, 3 fours, 1 fives, 3 sixes

θ_{MLE} :

$$p_1 = 3/12$$

$$p_2 = 2/12$$

$$p_3 = 0/12$$

$$p_4 = 3/12$$

$$p_5 = 1/12$$

$$p_6 = 3/12$$



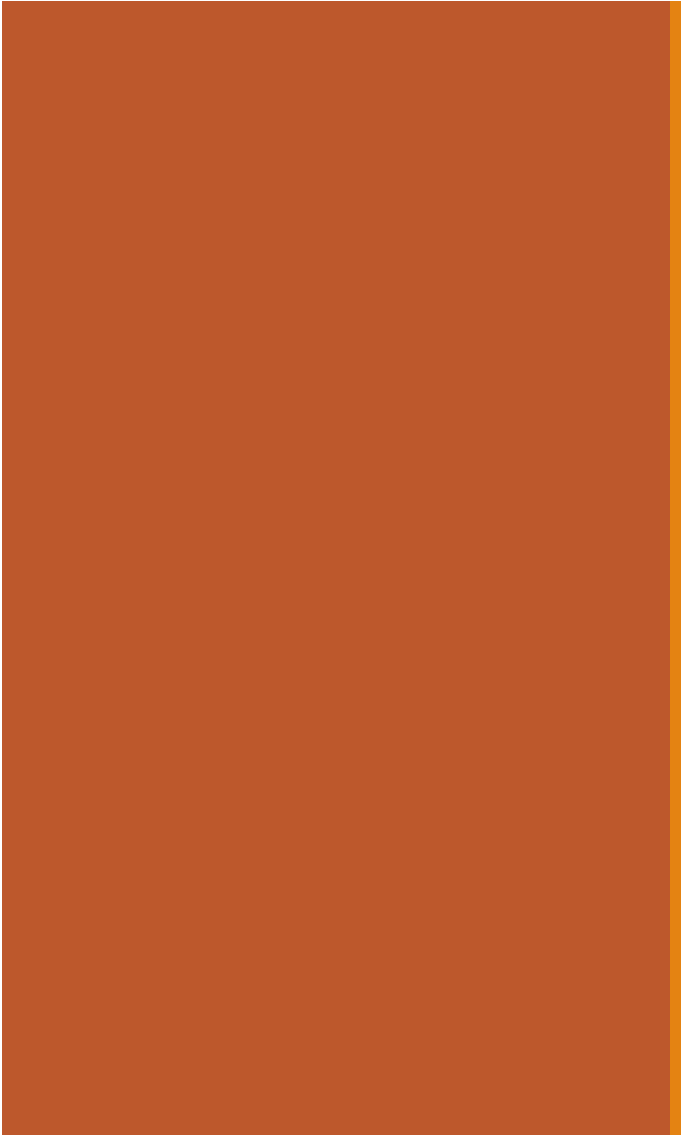
- MLE say you just never roll threes.
- Do you really believe that? *was 12 rolls really enough to discredit 3 as a possible outcome?*

Roll more!
prob = frequency
in limit

Frequentist

But what if you can't roll any more dice? More generally, what if you can't run enough experiments to realistically observe all possible outcomes.





Bayesian Statistics

Starting Today!

Today we are going to learn something unintuitive,
beautiful, and useful!

We are going to think of probabilities as
random variables.

A new definition of probability

Flip a coin $n + m$ times, produce n heads.

We don't know the **probability** θ that the coin comes up heads.



The world's first coin

Frequentist

θ is a single value.

$$\theta = \lim_{n+m \rightarrow \infty} \frac{n}{n+m} \approx \frac{n}{n+m}$$

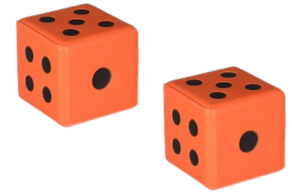
Bayesian

θ is a **random variable**.

θ 's continuous support: $(0, 1)$

Let's play a game

Roll 2 dice. If **neither** roll is a 6, you win (event W). Else, I win (event W^C).



- Before you play, what's the probability that you win?
- Play once. What's the probability that you win?
- Play three more times. What's the probability that you win?



Frequentist

$$P(W) = \left(\frac{5}{6}\right)^2$$



Bayesian

I am constantly re-evaluating the situation

Bayesian statistics: Constantly update your prior beliefs.

Bayesian probability

Bayesian statistics: Probability represents our ever-evolving understanding of the world.

Mixing discrete and continuous random variables, combined with Bayes' Theorem, allows us to reason about **probabilities as random variables**.

Mixing discrete and continuous

Let X be a continuous random variable, and N be a discrete random variable.

Bayes' Theorem:

$$f_{X|N}(x|n) = \frac{p_{N|X}(n|x)f_X(x)}{p_N(n)}$$

Intuition:
$$P(X = x|N = n) = \frac{P(N = n|X = x)P(X = x)}{P(N = n)}$$

$$P(X \approx x) = \int_{x-\epsilon_x/2}^{x+\epsilon_x/2} f_X(x) dx \approx f_X(x) \epsilon_x$$

$$f_{X|N}(x|n)\epsilon_x = \frac{P(N = n|X = x)f_X(x)\epsilon_x}{P(N = n)} \Rightarrow f_{X|N}(x|n) = \frac{p_{N|X}(n|x)f_X(x)}{p_N(n)}$$

Handwritten notes: "cancels" under ϵ_x in the numerator and denominator of the first equation.

Bayes' Theorem: All Flavors

Let X, Y be **continuous** and M, N be **discrete** random variables.

Original Bayes:

$$p_{M|N}(m|n) = \frac{p_{N|M}(n|m)p_M(m)}{p_N(n)}$$

} this is our original
Bayes' theorem
from Lecture 4

Mix Bayes #1:

$$f_{X|N}(x|n) = \frac{p_{N|X}(n|x)f_X(x)}{p_N(n)}$$

} from prior
slide

Mix Bayes #2:

$$p_{N|X}(n|x) = \frac{f_{X|N}(x|n)p_N(n)}{f_X(x)}$$

} another mixed form
not unlike the one
above it.

All continuous:

$$f_{X|Y}(x|y) = \frac{f_{Y|X}(y|x)f_X(x)}{f_Y(y)}$$

} from Lecture 17
purely continuous
form

Mixing discrete and continuous

Let θ be a random variable for the probability your coin comes up heads, and N be the number of heads you observe in an experiment.

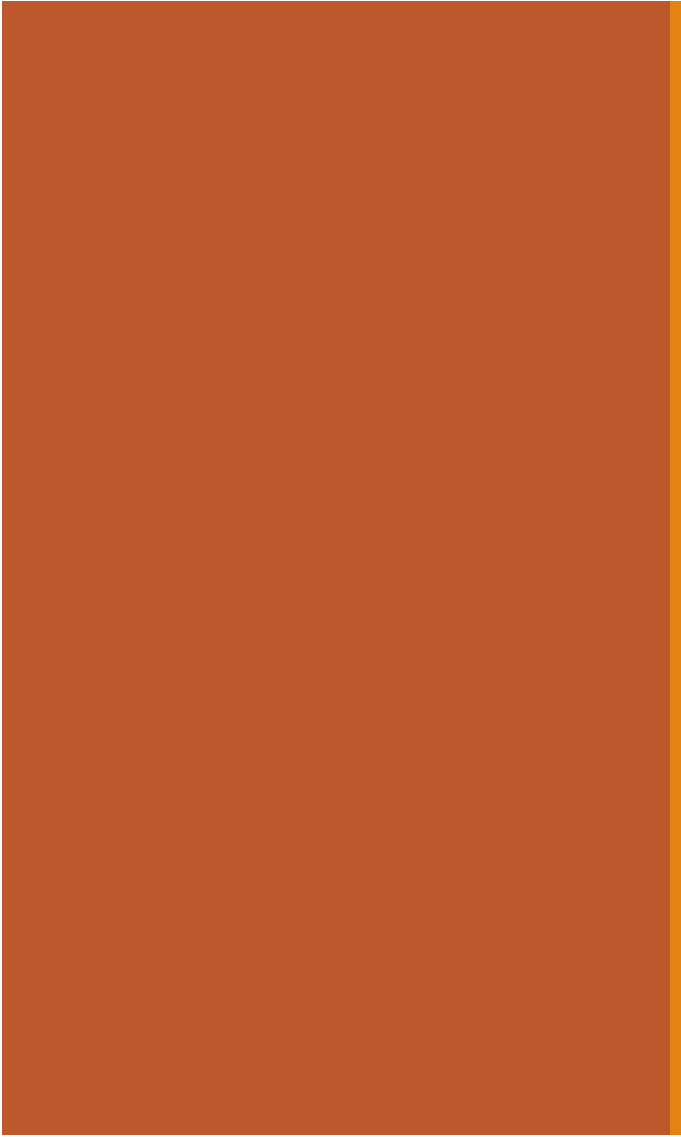
Assume $n+m$ trials where n heads come up.

$$\text{posterior } f_{\theta|N}(x|n) = \frac{\text{likelihood } p_{N|\theta}(n|x) \text{ prior } f_{\theta}(x)}{p_N(n)}$$

normalization constant

- **Prior** belief of parameter θ
- **Likelihood** of $N = n$ heads, given parameter $\theta = x$.
- **Posterior** updated belief of parameter θ .

$$\begin{aligned} f_{\theta}(x) \\ p_{N|\theta}(n|x) \\ f_{\theta|N}(x|n) \end{aligned}$$

A large, solid orange rectangle occupies the left side of the slide, extending from the top to the bottom and from the left edge to approximately one-third of the way across the slide.

Beta RV

Beta random variable

def A **Beta** random variable X is defined as follows:

$$X \sim \text{Beta}(a, b)$$

$$a > 0, b > 0$$

Support of X : $(0, 1)$

$$\text{PDF } \underline{f(x)} = \frac{1}{B(a, b)} x^{a-1} (1-x)^{b-1}$$

where $B(a, b) = \int_0^1 x^{a-1} (1-x)^{b-1} dx$, normalizing constant

$$\text{Expectation } E[X] = \frac{a}{a+b}$$

$$\text{Variance } \text{Var}(X) = \frac{ab}{(a+b)^2(a+b+1)}$$

$B(a, b)$ is chosen so that $\int_0^1 \underline{f(x)} dx = 1$

$$\frac{1}{B(a, b)} \int_0^1 x^{a-1} (1-x)^{b-1} dx = 1 \Rightarrow B(a, b) = \int_0^1 x^{a-1} (1-x)^{b-1} dx$$

Beta RV with different a, b

$$X \sim \text{Beta}(a, b)$$

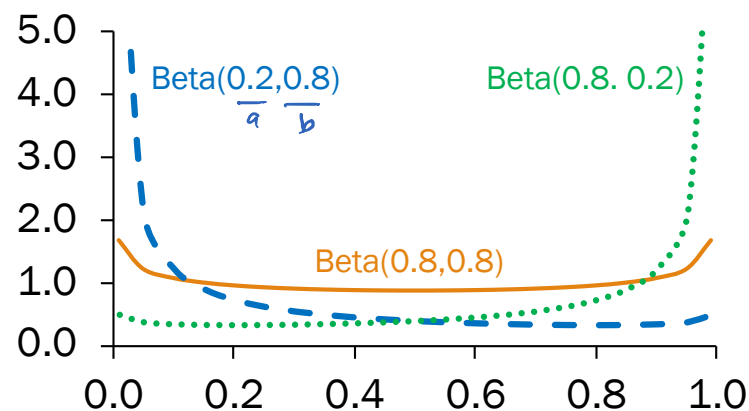
$$a > 0, b > 0$$

Support of X : $(0, 1)$

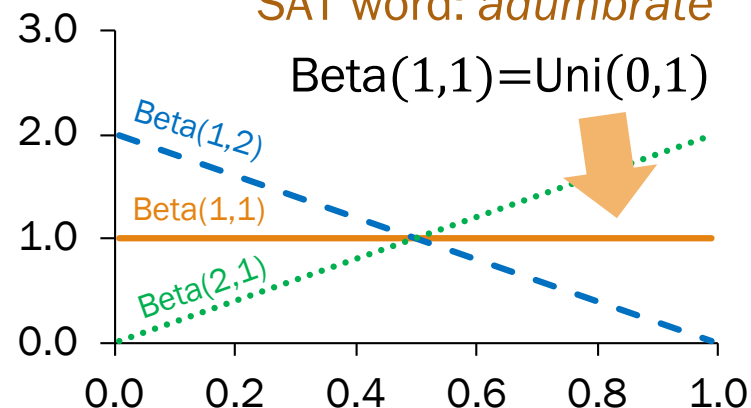
$$\text{PDF } f(x) = \frac{1}{B(a, b)} x^{a-1} (1-x)^{b-1}$$

where $B(a, b) = \int_0^1 x^{a-1} (1-x)^{b-1} dx$, normalizing constant

examples where $a, b < 1$



SAT word: *adumbrate*



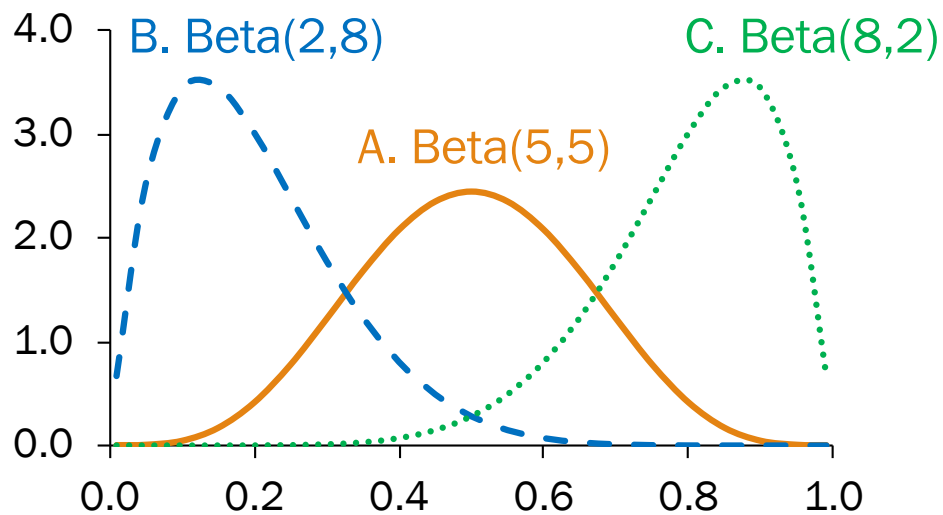
+ a third case
(next slide)

Note: PDF symmetric when $a = b$

Beta RV with different a, b

$$X \sim \text{Beta}(a, b)$$

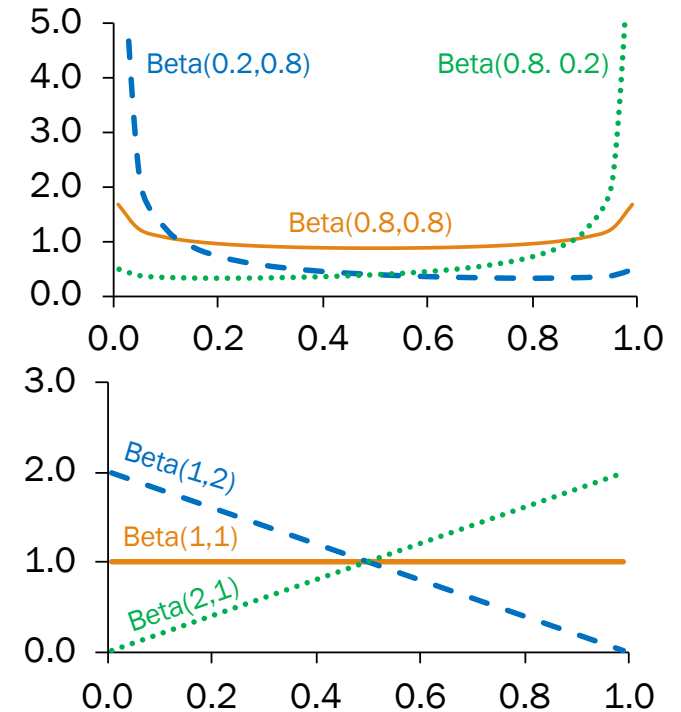
Match PDF to distribution:



*a, b
are integers
greater than 1*

- A. Beta(5,5)
- B. Beta(2,8)
- C. Beta(8,2)

In CS109, we focus on Beta functions where a, b are both positive integers.

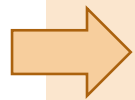


Beta random variable

def A **Beta** random variable X is defined as follows:

$$X \sim \text{Beta}(a, b)$$

$$a > 0, b > 0$$



Support of X : $(0, 1)$

PDF $f(x) = \frac{1}{B(a, b)} x^{a-1} (1-x)^{b-1}$

where $B(a, b) = \int_0^1 x^{a-1} (1-x)^{b-1} dx$, normalizing constant

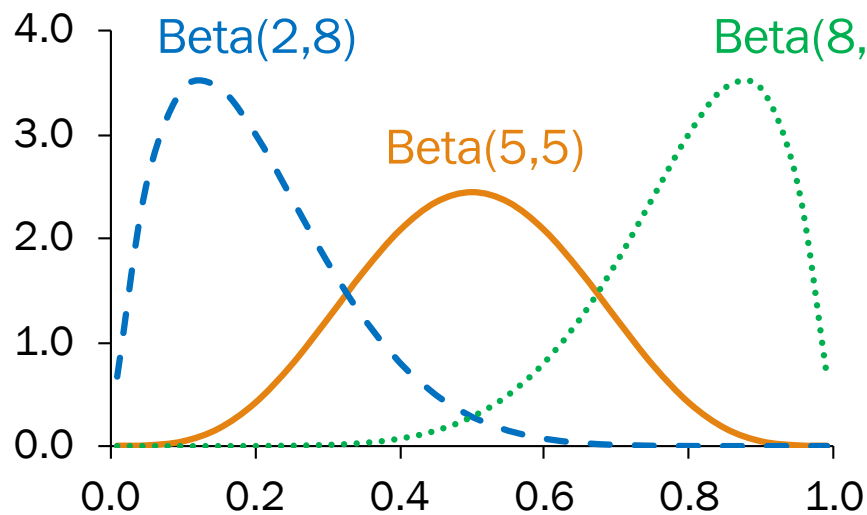
Expectation $E[X] = \frac{a}{a+b}$

Variance $\text{Var}(X) = \frac{ab}{(a+b)^2(a+b+1)}$

Beta can be a distribution of probabilities.

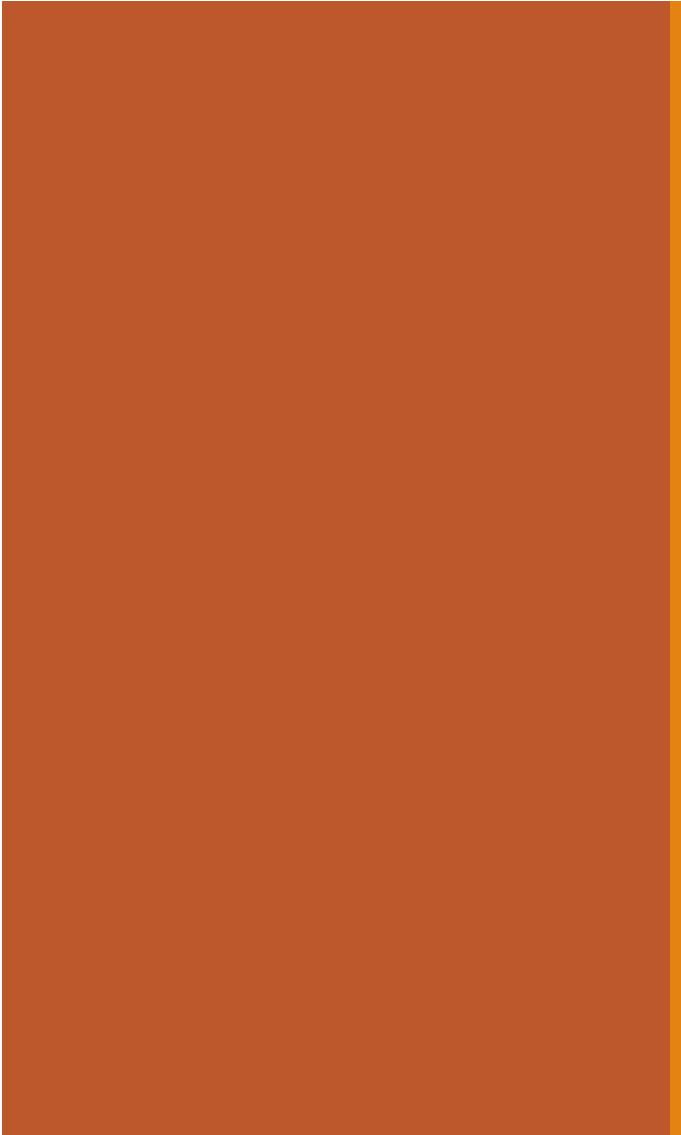
Beta can be a distribution of probabilities.

$$X \sim \text{Beta}(a, b)$$



Beta parameters a, b are determined by the outcome of an experiment.

But which experiment?



Flipping a coin with unknown probability

Flip a coin with unknown probability

Flip a coin $n + m$ times, observe n heads.

- Before our experiment, θ (the probability that the coin comes up heads) is equally like to be any probability in $(0, 1)$.
- Let N = number of heads.
- Given $\theta = x$, coin flips are independent.

What is our updated belief of θ after we observe $N = n$?

What are reasonable distributions of the following?

1. θ Bayesian prior $\theta \sim \text{Uni}(0,1)$
2. $N|\theta = x$ Likelihood $N|\theta = x \sim \text{Bin}(n + m, x)$
expression bracketed in blue is the random variable
3. $\theta|N = n$ Bayesian posterior. Use Bayes'!



Flip a coin with unknown probability

Flip a coin $n + m$ times, observe n heads.

- Before our experiment, θ (the probability that the coin comes up heads) is equally like to be any probability in $(0, 1)$.
- Let N = number of heads.
- Given $\theta = x$, coin flips are independent.

Prior:
 $\theta \sim \text{Uni}(0, 1)$

Likelihood:
 $N | \theta = x \sim \text{Bin}(n + m, x)$

What is our updated belief of θ after we observe $N = n$?

Posterior: $f_{\theta|N}(\theta|n)$

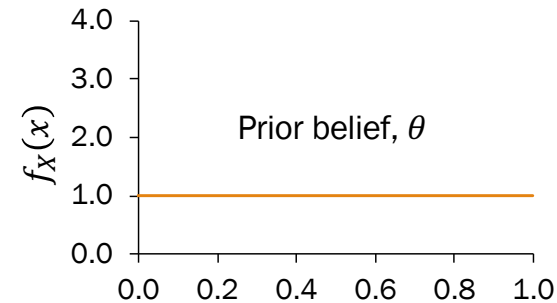
$$\begin{aligned} f_{\theta|N}(x|n) &= \frac{p_{N|\theta}(n|x)f_{\theta}(x)}{p_N(n)} = \frac{\binom{n+m}{n} x^n (1-x)^m \cdot 1}{p_N(n)} \\ &= \underbrace{\frac{\binom{n+m}{n}}{p_N(n)}}_{\text{constant with respect to } x} x^n (1-x)^m = \frac{1}{c} x^n (1-x)^m, \text{ where } c = \int_0^1 x^n (1-x)^m dx \end{aligned}$$

constant with respect to x ,
doesn't depend on x

Yi Ma, Yan, Chris Piech, Mehran Sahami, and Jerry Cain, CS109, Spring 2024

Let's try it out

1. Start with a $\theta \sim \text{Uni}(0,1)$ over probability that a coin lands heads.
2. Flip a coin 8 times. Observe $n = 7$ heads and $m = 1$ tail
3. What is our posterior belief of the probability θ ?



$$f_{\theta|N}(x|n) = \frac{1}{c} x^7 (1 - x)^1$$

c normalizes to valid PDF

Wait a minute! #looksbetalike

Beta RV with different a, b

$$X \sim \text{Beta}(a, b)$$

$$a > 0, b > 0$$

Support of X : $(0, 1)$

$$\text{PDF } f(x) = \frac{1}{B(a, b)} x^{a-1} (1-x)^{b-1}$$

where $B(a, b) = \int_0^1 x^{a-1} (1-x)^{b-1} dx$, normalizing constant



$$f_{\theta|N}(x|n) = \frac{1}{c} x^7 (1-x)^1 \quad \text{is the PDF for Beta}(8, 2)!$$

c normalizes to valid PDF

Let's try it out

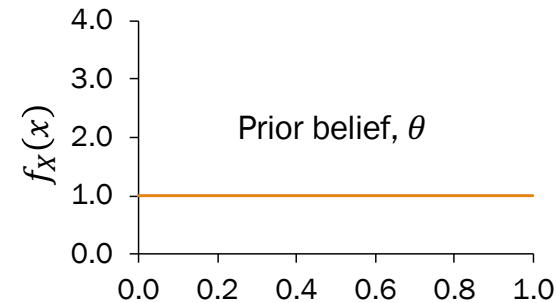
1. Start with a $\theta \sim \text{Uni}(0,1)$ over probability that a coin lands heads.

2. Flip a coin 8 times. Observe $n = 7$ heads and $m = 1$ tail

3. What is our posterior belief of the probability θ ?

neat!

Beta(8,2)



$$f_{\theta|N}(x|n) = \frac{1}{c} x^7 (1 - x)^1$$

c normalizes to valid PDF

3. What is our posterior belief of the probability θ ?

- Start with a $\theta \sim \text{Uni}(0,1)$ over probability
- Observe $n = 7$ successes and $m = 1$ failures
- Your new belief about the probability of θ is:

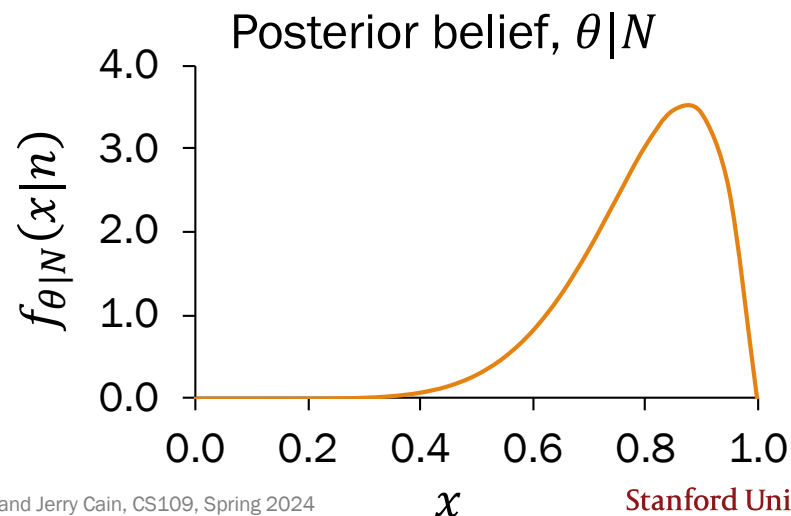
$$f_{\theta|N}(x|n) = \frac{1}{c} x^7 (1-x)^1, \text{ where } c = \int_0^1 x^7 (1-x)^1 dx$$

Posterior belief, $\theta|N$:

Beta($a = 8, b = 2$)

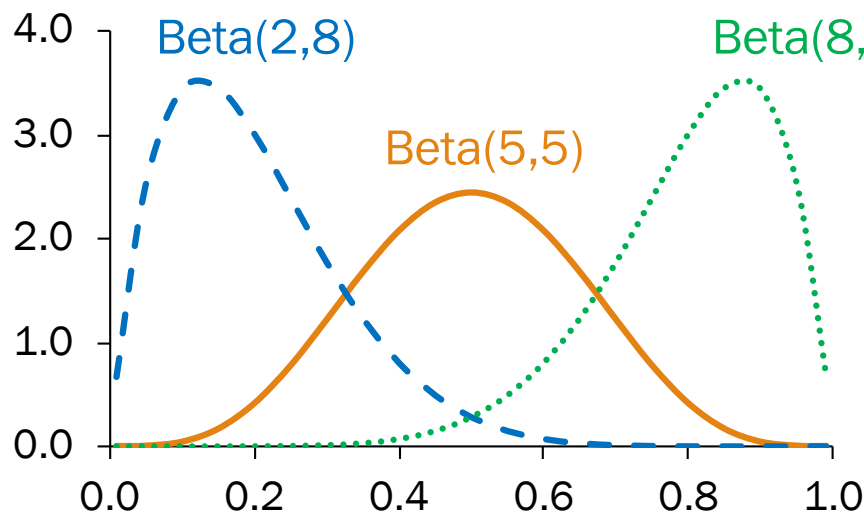
$$f_{\theta|N}(x|n) = \frac{1}{c} x^{8-1} (1-x)^{2-1}$$

$\downarrow$ # successes $\downarrow$ # failures
Beta($a = n + 1, b = m + 1$)



CS109 focus: Beta where a, b both positive integers

$X \sim \text{Beta}(a, b)$



Beta parameters a, b are determined by the outcome of an experiment.

$$a = \text{“successes”} + 1$$
$$b = \text{“failures”} + 1$$

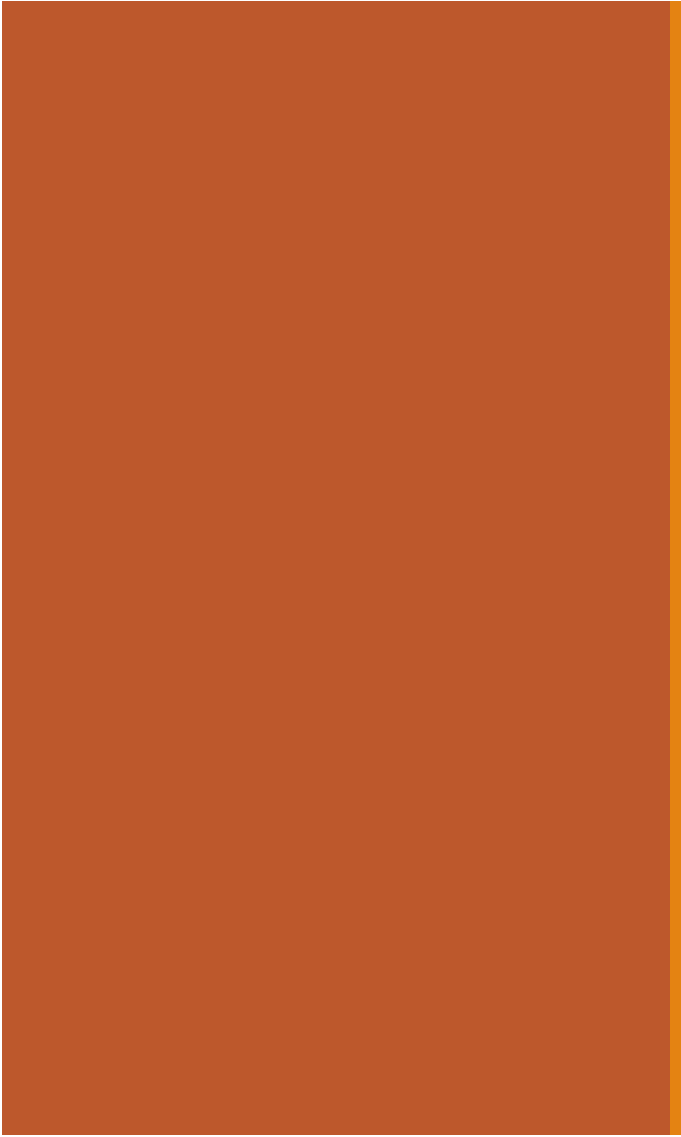
redux:

MLE: θ is a point estimate

Bayesian view: θ is a random variable

Beta: θ expresses the probability of a probability

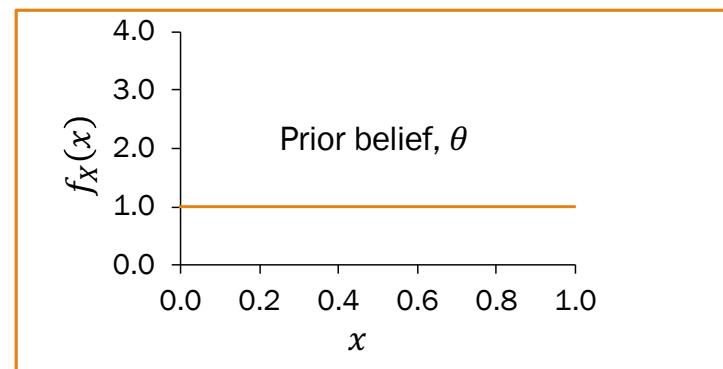
- Beta can model the randomness of the probability of experiment success.
- Beta parameters depend on our data and our prior.

A large orange rectangle with a thin vertical orange line on its right side, positioned on the left side of the slide.

Conjugate distributions

A note about our prior

1. Start with a $\theta \sim \text{Uni}(0,1)$ over probability that a coin lands heads.



2. Flip a coin 8 times. Observe $n = 7$ heads and $m = 1$ tail

okay 📱👍

3. What is our posterior belief of the probability θ ?

$$f_{\theta|N}(x|n) = \frac{1}{c} x^7 (1 - x)^1$$

c normalizes to valid PDF

Beta(8,2)

Wait another minute!

Beta RV with different a, b

Review

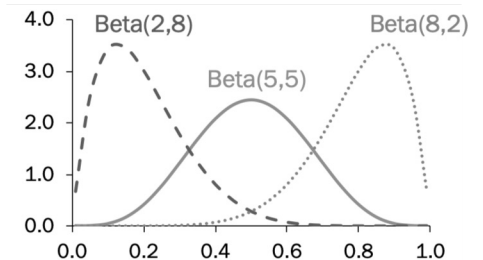
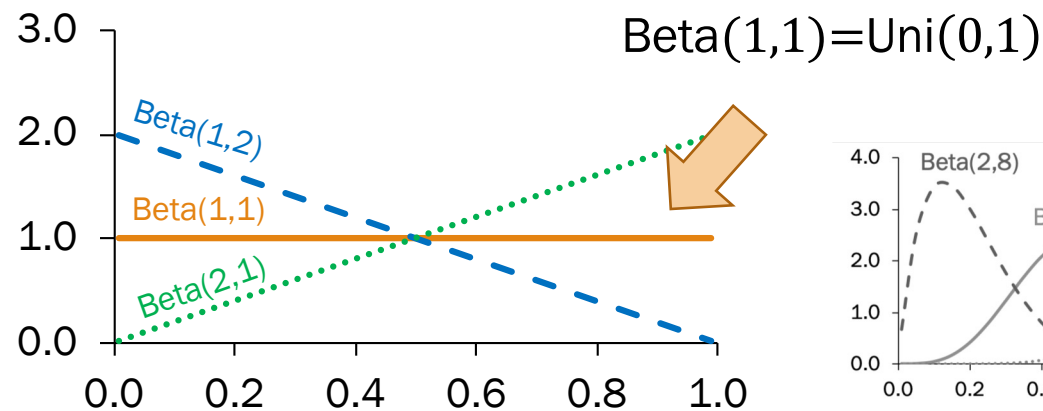
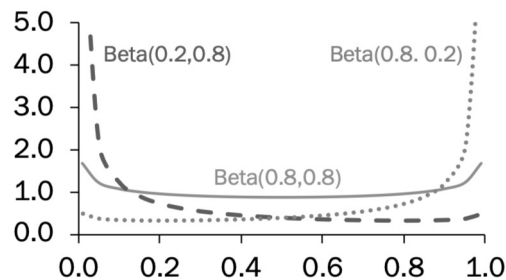
$$X \sim \text{Beta}(a, b)$$

$$a > 0, b > 0$$

Support of X : $(0, 1)$

$$\text{PDF } f(x) = \frac{1}{B(a, b)} x^{a-1} (1-x)^{b-1}$$

where $B(a, b) = \int_0^1 x^{a-1} (1-x)^{b-1} dx$, normalizing constant



Note: PDF symmetric when $a = b$

A note about our prior

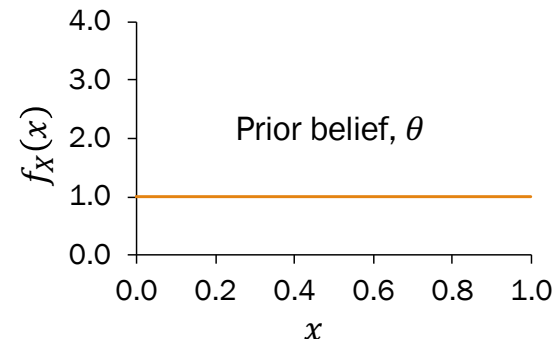
1. Start with a $\theta \sim \text{Uni}(0,1)$ over probability that a coin lands heads.

Beta(1,1)

2. Flip a coin 8 times. Observe $n = 7$ heads and $m = 1$ tail

3. What is our posterior belief of the probability θ ?

Beta(8,2)



Check this out. Beta($a = 1, b = 1$):

$$f(x) = \frac{1}{B(a, b)} x^{a-1} (1-x)^{b-1}$$

$$= \frac{1}{\int_0^1 1 dx}$$

$$= 1 \quad \text{where } 0 < x < 1$$

confirms Beta(1,1) is just Uni(0,1)

Beta is a conjugate distribution for Bernoulli

Beta is a **conjugate distribution** for Bernoulli, meaning:

- Prior and posterior parametric forms are the same

(proof on next slide)

Beta is a conjugate distribution for Bernoulli

Beta is a **conjugate distribution** for Bernoulli, meaning:

1. If our prior belief of the parameter is Beta, and
2. Our experiment is Bernoulli, then (observe n successes, m failures)
3. Our posterior is also Beta.

Proof: $\theta \sim \text{Beta}(a, b)$ $N|\theta \sim \text{Bin}(n + m, x)$

$$f_{\theta|N}(x|n) = \frac{p_{N|\theta}(n|x) f_{\theta}(x)}{p_N(n)} = \frac{\binom{n+m}{m} x^n (1-x)^m \cdot \frac{1}{B(a,b)} x^{a-1} (1-x)^{b-1}}{p_N(n)}$$

constants that
don't depend on x

$$= C \cdot x^n (1-x)^m \cdot x^{a-1} (1-x)^{b-1}$$

$$= C \cdot x^{n+a-1} (1-x)^{m+b-1} \quad \checkmark \Rightarrow \text{Beta}(a,b) \text{ prior evolved into Beta}(a+n, b+m) \text{ posterior}$$

Beta is a conjugate distribution for Bernoulli

This is the main
takeaway of Beta.

Beta is a **conjugate distribution** for Bernoulli, meaning:

- Prior and posterior parametric forms are the same $\rightarrow$ both are Beta
- Practically, conjugate means easy update:
Add number of “heads” and “tails” seen to Beta parameters.

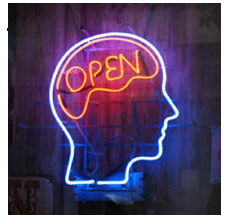
You can **invent a prior** to express how biased you believe the coin is a priori:

- $\theta \sim \text{Beta}(a, b)$: pretend you’ve conducted $(a + b - 2)$ **imaginary trials**, where $(a - 1)$ trials produced a head and $(b - 1)$ produced a tail
- Choosing $\text{Beta}(1, 1) = \text{Uni}(0, 1)$ means you don’t hold any prior beliefs

Prior $\text{Beta}(a = n_{\text{imag}} + 1, b = m_{\text{imag}} + 1)$

Experiment Observe n successes and m failures $\leftarrow$ inherently Binomial

Posterior $\text{Beta}(a = n_{\text{imag}} + n + 1, b = m_{\text{imag}} + m + 1)$



Medicinal Beta

- Before being tested, a medicine is believed to "work" 80% of the time.
- The medicine is administered to 20 patients.
- It "works" for 14, "doesn't work" for 6.

What is your new belief that the drug "works"?

Frequentist

Let p be the probability
your drug works.

$$p \approx \frac{14}{20} = 0.7$$

*viewpoint: prior beliefs irrelevant, go
with most recent data*

Bayesian

A frequentist view will not incorporate
prior/expert belief about probability.

Medicinal Beta

- Before being tested, a medicine is believed to "work" 80% of the time.
- The medicine is administered to 20 patients.
- It "works" for 14, "doesn't work" for 6.

What is your new belief that the drug "works"?

Frequentist

Let p be the probability
your drug works.

$$p \approx \frac{14}{20} = 0.7$$

Bayesian

Let θ be the probability
your drug works.

θ is a random variable.
*viewpoint: history matters and
shouldn't be ignored.*

Medicinal Beta

Prior $\text{Beta}(a = n_{\text{imag}} + 1, b = m_{\text{imag}} + 1)$

Posterior $\text{Beta}(a = n_{\text{imag}} + n + 1, b = m_{\text{imag}} + m + 1)$

- Before being tested, a medicine is believed to "work" 80% of the time.
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*when using Beta to capture prior beliefs,
we must invent values of a and b
consistent with the 80%*

What is your new belief that the drug "works"?

(Bayesian interpretation)

What is the prior distribution of θ ? (select all that apply)

- A. $\theta \sim \text{Beta}(1, 1) = \text{Uni}(0, 1)$
- B. $\theta \sim \text{Beta}(81, 101)$
- C. $\theta \sim \text{Beta}(80, 20)$
- D. $\theta \sim \text{Beta}(81, 21)$
- E. $\theta \sim \text{Beta}(5, 2)$



Medicinal Beta

Prior	$\text{Beta}(a = n_{\text{imag}} + 1, b = m_{\text{imag}} + 1)$
Posterior	$\text{Beta}(a = n_{\text{imag}} + n + 1, b = m_{\text{imag}} + m + 1)$

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- B. $\theta \sim \text{Beta}(81, 101)$
- C. $\theta \sim \text{Beta}(80, 20)$
- ☒ D. $\theta \sim \text{Beta}(81, 21) \rightarrow$ Interpretation: 80 successes / 100 imaginary trials
- ☒ E. $\theta \sim \text{Beta}(5, 2) \rightarrow$ 4 successes / 5 imaginary trials

(you can choose either based on how strongly you believe in prior data.)

We choose E on next slide)

Medicinal Beta

Prior $\text{Beta}(a = n_{\text{imag}} + 1, b = m_{\text{imag}} + 1)$

Posterior $\text{Beta}(a = n_{\text{imag}} + n + 1, b = m_{\text{imag}} + m + 1)$

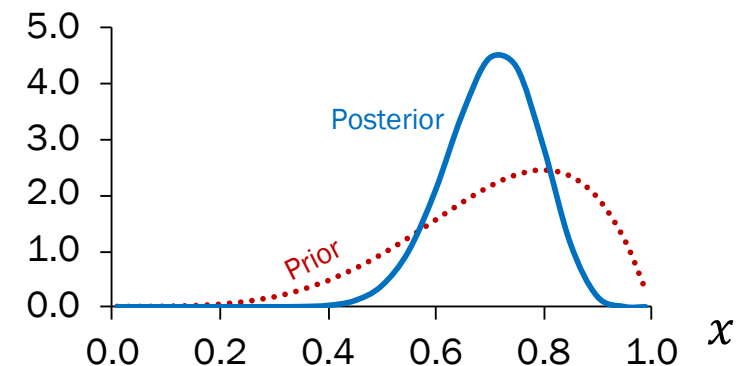
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What is your new belief that the drug "works"?

(Bayesian interpretation)

Prior: $\theta \sim \text{Beta}(a = 5, b = 2)$

Posterior: $\theta \sim \text{Beta}(a = 5 + 14, b = 2 + 6)$
 $\sim \text{Beta}(a = 19, b = 8)$



Medicinal Beta

Prior	$\text{Beta}(a = n_{\text{imag}} + 1, b = m_{\text{imag}} + 1)$
Posterior	$\text{Beta}(a = n_{\text{imag}} + n + 1, b = m_{\text{imag}} + m + 1)$

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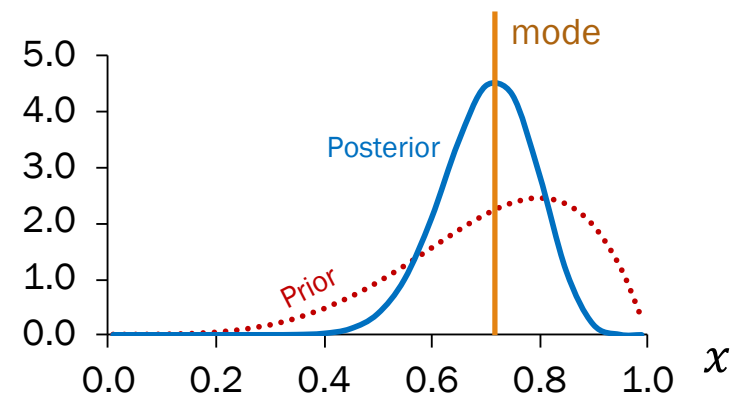
Prior: $\theta \sim \text{Beta}(a = 5, b = 2)$

Posterior: $\theta \sim \text{Beta}(a = 5 + 14, b = 2 + 6)$
 $\sim \text{Beta}(a = 19, b = 8)$

What do you report to pharmacists?

- A. Expectation of posterior
- B. Mode of posterior
- C. Distribution of posterior
- D. Nothing

(Bayesian interpretation)



Medicinal Beta

Prior $\text{Beta}(a = n_{\text{imag}} + 1, b = m_{\text{imag}} + 1)$

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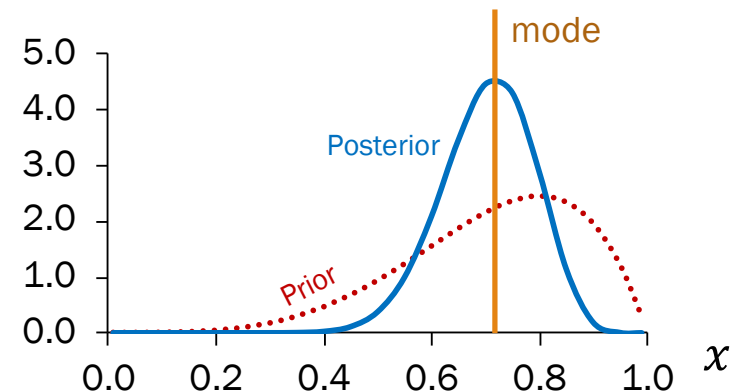
What do you report to pharmacists?

$$E[\theta] = \frac{a}{a+b} = \frac{19}{19+8} \approx 0.70$$

$$\text{mode}(\theta) = \frac{a-1}{a+b-2} = \frac{18}{18+7} \approx 0.72$$

will prove later 😊

(Bayesian interpretation)



In CS109, we report the **mode**: The "most likely" parameter given the data.

Food for thought



In this lecture:

$$X \sim \text{Ber}(p)$$

If nothing is known about the **parameter** p , Bayesian statisticians will:

- Treat the parameter as a random variable θ with a Beta prior distribution
- Conduct experiments
- Based on the outcomes of those experiments, update the posterior distribution of θ

Food for thought:

Any parameter for a “parameterized” random variable can be thought of as a random variable.

$$Y \sim \mathcal{N}(\mu, \sigma^2)$$

Estimating our parameter directly

(our focus so far)

Maximum
Likelihood
Estimator
(MLE)

What is the parameter θ
that **maximizes the
likelihood**
of our observed data
($x_1, x_2, \dots, x_n$)?

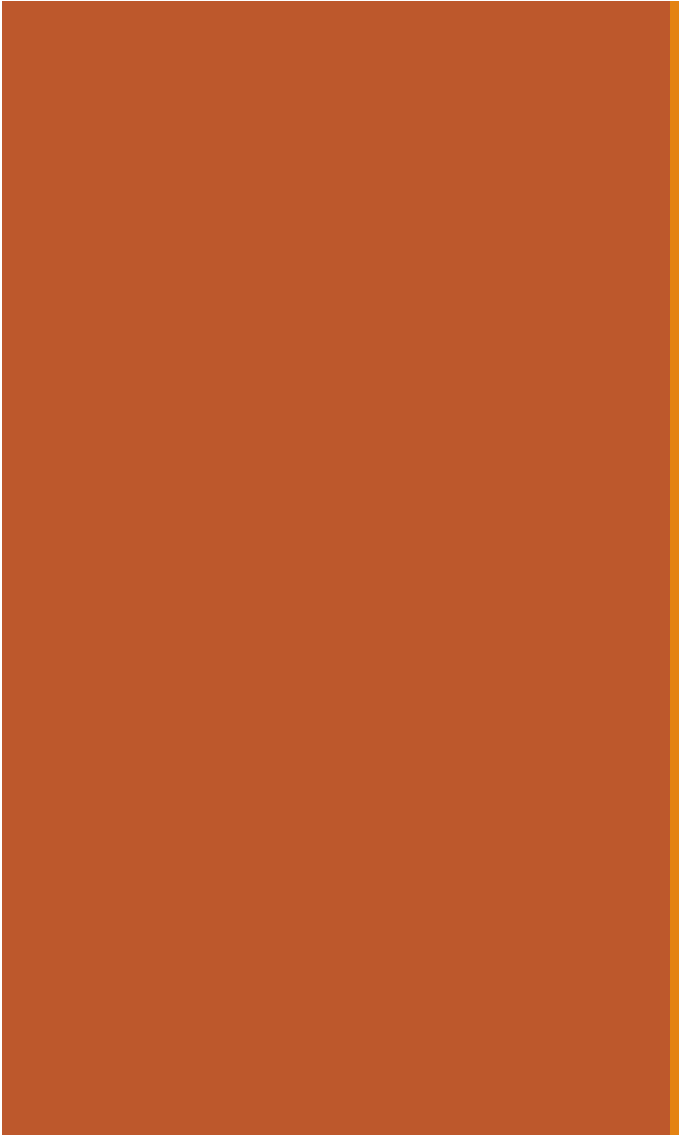
$$L(\theta) = f(X_1, X_2, \dots, X_n | \theta) \\ = \prod_{i=1}^n f(X_i | \theta)$$

$$\theta_{MLE} = \arg \max_{\theta} f(X_1, X_2, \dots, X_n | \theta)$$

likelihood of data

Observations:

- MLE maximizes probability of observing data given a parameter θ . It's fitting the curve to match the data.
- If we are estimating θ , shouldn't we **maximize the probability of θ** directly? SAT word: *adumbrate*



Extra: MLE: Multinomial derivation

Okay, just one more MLE with the Multinomial

Consider a sample of n i.i.d. random variables where

- Each element is drawn from one of m outcomes.
 $P(\text{outcome } i) = p_i$, where $\sum_{i=1}^m p_i = 1$
- $X_i = \#$ of trials with outcome i , where $\sum_{i=1}^m X_i = n$

1. What is the likelihood of observing the sample $(X_1, X_2, \dots, X_m)$, given the probabilities $p_1, p_2, \dots, p_m$?

$$L(\theta) = \frac{n!}{X_1! X_2! \cdots X_m!} p_1^{X_1} p_2^{X_2} \cdots p_m^{X_m}$$

2. What is θ_{MLE} ?

$$LL(\theta) = \log(n!) - \sum_{i=1}^m \log(X_i!) + \sum_{i=1}^m X_i \log(p_i), \text{ such that } \sum_{i=1}^m p_i = 1$$

$$\theta_{MLE}: p_i = \frac{X_i}{n} \quad \text{Intuitively, probability } p_i = \text{proportion of outcomes}$$

Optimizing MLE for Multinomial

$$\theta = (p_1, p_2, \dots, p_m)$$

$$\theta_{MLE} = \arg \max_{\theta} LL(\theta), \text{ where } \sum_{i=1}^m p_i = 1$$

Use Lagrange multipliers
to account for constraint

Lagrange multipliers:

$$A(\theta) = LL(\theta) + \lambda \left(\sum_{i=1}^m p_i - 1 \right) = \sum_{i=1}^m X_i \log(p_i) + \lambda \left(\sum_{i=1}^m p_i - 1 \right) \quad (\text{drop non-}p_i \text{ terms})$$

Differentiate w.r.t. each p_i , in turn:

$$\frac{\partial A(\theta)}{\partial p_i} = X_i \frac{1}{p_i} + \lambda = 0 \Rightarrow p_i = -\frac{X_i}{\lambda}$$

Solve for λ , noting $\sum_{i=1}^m X_i = n, \sum_{i=1}^m p_i = 1$:

$$\sum_{i=1}^m p_i = \sum_{i=1}^m -\frac{X_i}{\lambda} = 1 \Rightarrow 1 = -\frac{n}{\lambda} \Rightarrow \lambda = -n$$

Substitute λ into p_i

$$p_i = \frac{X_i}{n}$$